UNLOX

MINOR PROJECT ASSESSMENT

Cyber Security and Networking Fundamentals

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Section A — Cybersecurity Fundamentals (15 Marks)

Q1. Explain cybersecurity and why its importance. Explain the CIA Triad with one practical example for each component. (5 Marks)

ANS:

- Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorized access, attacks, damage, or theft.
- Importance of cybersecurity:
- Protects personal and organizational information.
- Prevents unauthorized access and data theft.
- Maintains privacy and confidentiality.
- Reduces financial losses caused by cyberattacks.
- Ensures systems and services remain available.

CIA Triad

1. Confidentiality

Ensures that information is accessible only to authorized people.

Example: Using passwords and encryption to protect student records.



2. Integrity

Information must remain accurate and must not be changed without authorization.

Example: Preventing unauthorized changes to examination marks.



3. Availability

Systems and information must be accessible to authorized users when required.

Example: Maintaining backup servers so an online banking service can continue operating during a server failure.



Q2.Differentiate between vulnerability, threat, risk, and attack. Give one simple example for each.

ANS:

Term	Meaning	Example
Vulnerability	A weakness in a system.	Outdated software
Threat	Something that could cause harm.	A hacker attempting to Steal data.
Risk	The possibility and impact of harm.	Data may be stolen because software is outdated.
Attack	An actual attempt to exploit a weakness .	An attacker exploits a software vulnerability.

In simple words: A vulnerability is a weakness, a threat is a potential danger, risk is the possibility and impact of damage, and an attack is an attempt to cause harm.

Q3.Briefly explain any five areas of cybersecurity, such as network,application,cloud,endpoint,information or physical security.

ANS:

1. Network security: Protects networks from unauthorized access and attacks using firewalls and monitoring tools.

2. Application security: Protects software applications from vulnerabilities through secure coding, testing, and updates.

3. Cloud security: Protects data and applications stored on cloud platforms using access controls and encryption.

4. Endpoint security: Protects devices such as laptops, computers, and smartphones using antivirus software and endpoint protection tools.

5. Information security: Protects information against unauthorized access, alteration, disclosure, and destruction.

Other areas include physical security, which protects buildings and hardware, and identity security, which manages user authentication and access.

Section B — Networking Fundamentals (20 Marks)

Q4. What is networking? Define a computer network. Explain why organization use network and describe LAN, MAN, WAN, and PAN with one example each.

ANS:

A computer network is a group of interconnected computers and devices that communicate and share resources such as files, printers, and internet access.

Why organizations use networks:

- To share files and hardware.
- To communicate quickly.
- To access the internet.
- To centralize data and backups.
- To reduce operational costs.

Network type	Full form	Coverage
PAN	Personal Area Network	A few metres
LAN	Local Area Network	Room, building, or campus
MAN	Metropolitan Area Network	City or metropolitan area
WAN	Wide Area Network	Large geographical Areas

Q5. Explain the purpose hub, switch, router, firewall, and access point. Mention one major difference between a switch and a router.

ANS:

1. Hub

Connects multiple devices in a network and sends incoming data to all its ports.



2. Switch

Connects devices in a LAN and forwards frames to the appropriate port using MAC addresses.



3. Router

Connects different IP networks and forwards packets toward their destinations.



4. Firewall

Monitors and filters network traffic according to security rules.



5. Access Point

Allows wireless devices to connect to a network through Wi-Fi.



Major difference between switch and router: A switch mainly connects devices within the same LAN, whereas a router connects different IP networks.

Q6. Explain what happens when a user enters example.com in a browser. Include DNS,IP addressing,gateway/router,TCP/IP and the web server.Draw a simple flow diagram.

ANS:

When a user enters example.com, the following steps occur:

1.URL entry: The user types example.com into the browser.

2.DNS lookup: The Domain Name System (DNS) finds the IP address associated with the domain name, often using a cached result or a DNS server.

3.IP addressing: The browser uses the destination IP address to identify the web server.

4.Gateway/router: If the destination is outside the local network, the computer sends the packet to its default gateway, which forwards it toward the internet.

5.TCP/IP communication: IP handles packet addressing and routing. For HTTPS, TCP is commonly used to establish a connection, although HTTP/3 uses QUIC over UDP.

6.Secure connection: For HTTPS, the browser establishes a secure connection using TLS.

7.Web server: The browser sends an HTTP request to the web server.

8.Server response: The server returns a response containing HTML, CSS, JavaScript, images, or other resources.

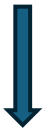
9.Rendering: The browser processes these resources and displays the website.

Flow diagram

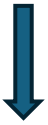
User enters example.com



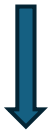
DNS resolves the domain to an IP address



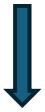
Computer → Default gateway / router



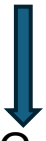
Internet routing using IP



Web server receives HTTP/HTTPS request



Response returns → Browser displays website



Section C — IP Addressing (20 Marks)

Q7. Differentiate IPv4 and IPv6. Explain why IPv6 was introduced and give one example address for each.

ANS:

An IP address is a logical address used to identify a device on an IP network.

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Address format	Decimal numbers separated by dots	Hexadecimal groups separated by colons
Address length	4 octets	8 groups of 16 bits
Address space	About 4.3 billion addresses	2^128 addresses
Example	192.168.1.10	2001:db8::1
Header	Variable length	Fixed base header of 40 bytes

Why was IPv6 introduced?

IPv6 was introduced mainly because IPv4 addresses were becoming insufficient due to the growth of internet-connected devices. It provides a vastly larger address space, simpler base-header processing, and improved support for network configuration.

Q8.IP Classification for 10.10.10.5,172.16.5.20,192.168.1.50 and 8.8.8.8, Identify the IPv4 class and private/public status where applicable.

ANS:

IPv4 class is identified by the first octet in traditional classful addressing.

IP address	First octet	Class	Private/Public
10.10.10.5	10	Class A	Private
172.16.5.20	172	Class B	Private
192.168.1.50	192	Class C	Private
8.8.8.8	8	Class A	Public

Explanation:

- Private Class A range: 10.0.0.0 to 10.255.255.255.
- Private Class B range: 172.16.0.0 to 172.31.255.255.
- Private Class C range: 192.168.0.0 to 192.168.255.255.
- 8.8.8.8 is a public IP address used by Google Public DNS.

Q9. For all IP components 192.168.20.45/24, identify network address,host portion,subnet mask,broadcost address and usable host range.show your working.

ANS:

Given:

IP address: 192.168.20.45/24

Prefix length: 24 bits

Step 1: Find the subnet mask

A /24 prefix means that the first 24 bits are network bits.

Subnet mask: 255.255.255.0

Step 2: Find the network address

IP address: 192.168.20.45

Subnet mask: 255.255.255.0

Performing the bitwise AND operation gives:

Network address = 192.168.20.0

Step 3: Find the host portion

The last 8 bits are reserved for the host portion.

Host portion = 45 (binary 00101101 in the last octet).

Step 4: Find the broadcast address

All host bits are set to 1.

Broadcast address = 192.168.20.255

Step 5: Find the usable host range

First usable address: 192.168.20.1

Last usable address: 192.168.20.254Step 6: Calculate the number of usable hosts

Host bits = 32-24=8

Total addresses = 2^8=256

Usable hosts = 256-2=254

Final answer

Network address 192.168.20.0

Host portion	45 in the last octet
Subnet mask	255.255.255.0
Broadcast address	192.168.20.255
Usable host range	192.168.20.1 – 192.168.20.254
Usable hosts	254

Section D — Subnetting & Calculations (25 Marks)

Q10. Basic subnet calculation for 192.168.10.0/26.

ANS:

Given: 192.168.10.0/26

Step 1: Calculate the subnet mask

A /26 prefix contains 26 network bits.

Subnet mask = 255.255.255.192

Step 2: Calculate the number of /26 subnets from /24

Borrowed bits = 26-24=2

Number of subnets:

2^2=4

Therefore, four/26 subnets can be formed from a/24 network.

Step 3: Calculate usable hosts per subnet

Host bits: 32-26=6

Total addresses: 2^6=64

Usable hosts: 64-2=62

Step 4: Calculate the block size

Block size = 256-192=64

The subnets increment by 64 in the last octet.

Subnet	Network address	First usable	Last usable	Broadcast
1	192.168.10.0/26	.1	.62	.63
2	192.168.10.64/26	.65	.126	.127
3	192.168.10.128/26	.129	.190	.191
4	192.168.10.192/26	.193	.254	.255

All shortened addresses in the table have the prefix 192.168.10.

Final answer for the first subnet, 192.168.10.0/26:

- Subnet mask: 255.255.255.192
- Number of subnets from /24: 4
- Usable hosts per subnet: 62
- Network address: 192.168.10.0
- First usable: 192.168.10.1
- Last usable: 192.168.10.62
- Broadcast address: 192.168.10.63

Q11. Design a subnetting scheme for IT, HR, Finance, and Management.

An organization has 192.168.10.0/24 and requires separate networks for IT—50 hosts, HR—30, Finance—20, Management—10. Design an appropriate subnetting scheme. Give Network, CIDR, Mask,

Usable Range, Broadcast and Host Count.

ANS:

Given network: 192.168.10.0/24

Requirements:

IT — 50 hosts
HR — 30 hosts
Finance — 20 hosts
Management — 10 hosts
We use VLSM (Variable Length Subnet Masking) to allocate subnet sizes according to the number of hosts required.
Step 1: Calculate the required subnet sizes

Department	Required hosts	Minimum host bits	CIDR	Usable hosts
IT	50	6	/26	62
HR	30	5	/27	30
Finance	20	5	/27	30
Management	10	4	/28	14

We allocate the largest subnet first and then assign the remaining address blocks consecutively.

Step 2: Assign the subnets

1. IT department

50 hosts required

Network: 192.168.10.0/26

Mask: 255.255.255.192

Usable range: 192.168.10.1 – 192.168.10.62

Broadcast: 192.168.10.63

Host capacity: 62 usable hosts

2. HR department

30 hosts required

Network: 192.168.10.64/27

Mask: 255.255.255.224

Usable range: 192.168.10.65 – 192.168.10.94

Broadcast: 192.168.10.95

Host capacity: 30 usable hosts

Finance department

20 hosts required

Network: 192.168.10.96/27

Mask: 255.255.255.224

Usable range: 192.168.10.97 – 192.168.10.126

Broadcast: 192.168.10.127

Host capacity: 30 usable hosts

4. Management department

10 hosts required

Network: 192.168.10.128/28

Mask: 255.255.255.240

Usable range: 192.168.10.129 – 192.168.10.142

Broadcast: 192.168.10.143

Host capacity: 14 usable hosts

Step 3: Verify the allocation

IT uses addresses .0–.63.

HR uses addresses .64–.95.

Finance uses addresses .96–.127.

Management uses addresses .128–.143.

No allocated subnets overlap.

Addresses .144–.255 remain available for future expansion.

Conclusion: VLSM efficiently divides the /24 network into four separate networks, each with sufficient host capacity for the department.

Section E — Scenario-Based Questions (20 Marks)

Q12. Scenario — Company Network

A company has 45 employees, 5 servers and 10 network devices. Design a basic network. State network

type, required devices, role of router/switch/firewall, and explain how an employee's request reaches a

website.

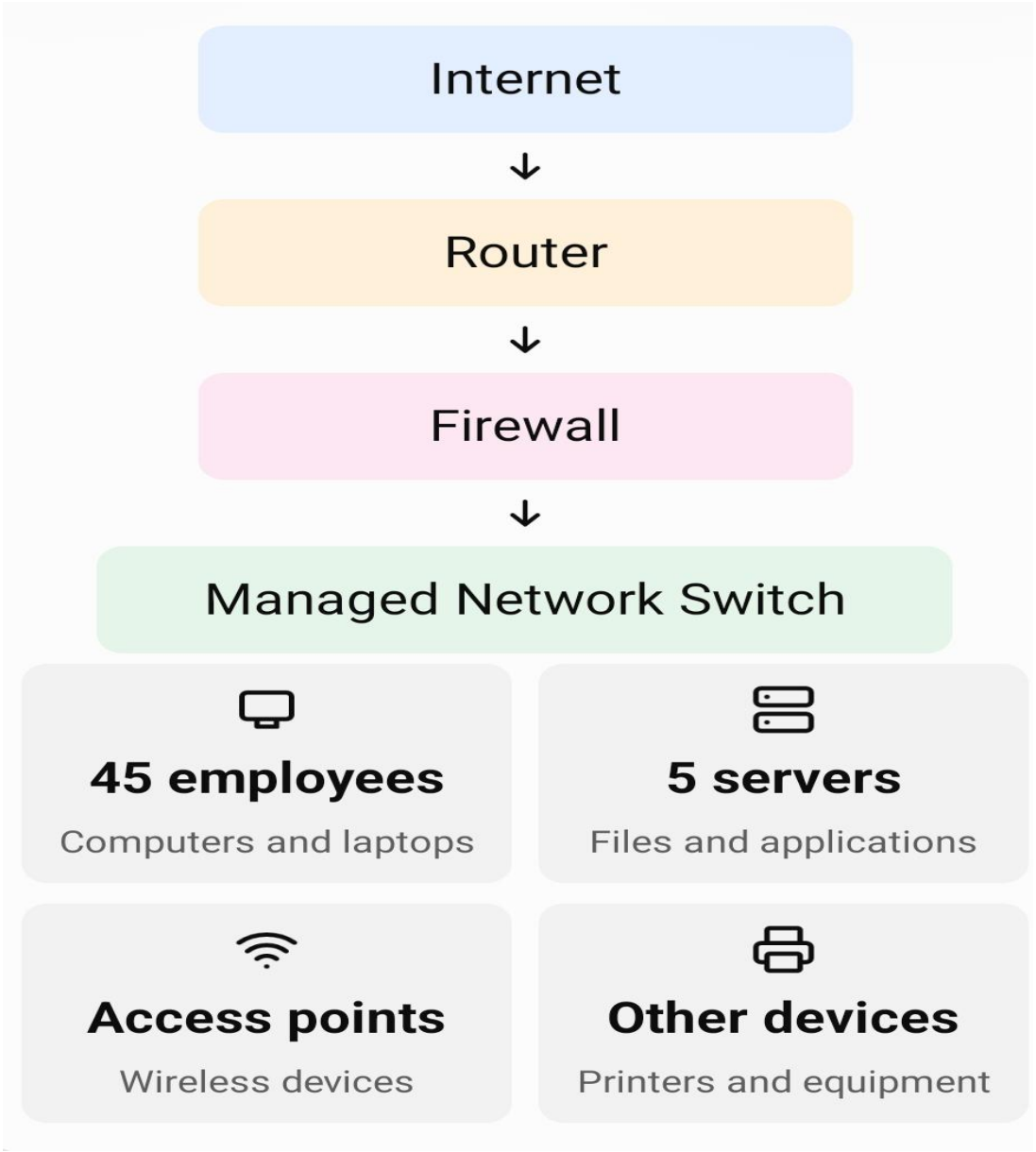
1. Network type

A LAN (Local Area Network) is suitable because the employees, servers, and network devices are located within the company premises.

2. Required network devices

- Router — connects the company LAN to the internet.
- Managed switch or switches — connect employee computers, servers, printers, and other wired devices.
- Firewall — protects the internal network from unauthorized traffic.
- Wireless access points — provide Wi-Fi connectivity.
- Servers — provide services such as file storage, applications, and authentication.
- Ethernet cables — connect wired devices.
- UPS — provides temporary power during power interruptions.

3.Network Diagram



4.Roles of the router,switch,and firewall

- Router: Routes packets between the company network and external networks.
- Switch: Connects devices within the company LAN and forwards frames using

MAC addresses.

- Firewall: Filters incoming and outgoing traffic according to security rules.

5.How an employee accesses a website

- The employee enters a website address in a browser.
- DNS resolves the domain name into an IP address.
- The computer sends packets through the local switch.
- The firewall checks the traffic against its security rules.
- The router forwards permitted traffic toward the internet.
- The website server sends a response back through the network.
- The browser displays the website.

Conclusion: A properly configured LAN with a router, firewall, switches, and access points enables secure communication, resource sharing, and internet access for company employees.

Q13. IP troubleshooting: A user has 192.168.1.25/24 and can communicate with another local computer but cannot access an external website. As a junior network engineer,list checks in order,including IP configuration,gateway,DNS,connectivity testing and router role.

ANS:

The user can communicate with another local computer, suggesting that local network connectivity is working. The problem may be related to the default gateway, router, DNS, or internet connection.

Step 1: Check IP configuration

Verify the following settings:

IP address: 192.168.1.25

Subnet mask: 255.255.255.0

Default gateway: The correct IP address of the local router, often 192.168.1.1 (verify the actual configuration).

DNS server: A valid and reachable DNS server.

On Windows, run:

ipconfig /all

Step 2: Check local connectivity

Ping another computer on the same network:

Replace the example address with the actual local computer's IP address. A successful response indicates local IP connectivity to that host.

ping 192.168.1.26

Step 3: Check the default gateway

Ping the configured gateway. For example:

If it fails, verify the gateway address, cable or Wi-Fi connection, VLAN settings, and router interface. Some devices block ping, so a failed response alone does not prove the gateway is down.

Step 4: Check external IP connectivity

Test connectivity to a known external IP address:

ping 8.8.8.8

If it succeeds, external IP connectivity is likely working.

If it fails, check the router's internet connection, routing, firewall rules, and upstream service. Note that some external hosts may block ping.

Step 5: Check DNS resolution

Run:

nslookup example.com

If an IP address is returned, DNS resolution is working.

If resolution fails, verify DNS server settings and DNS connectivity.

If appropriate, clear the local DNS cache:

ipconfig /flushdns

Step 6: Test the website using HTTPS

Open the website in a browser or use:

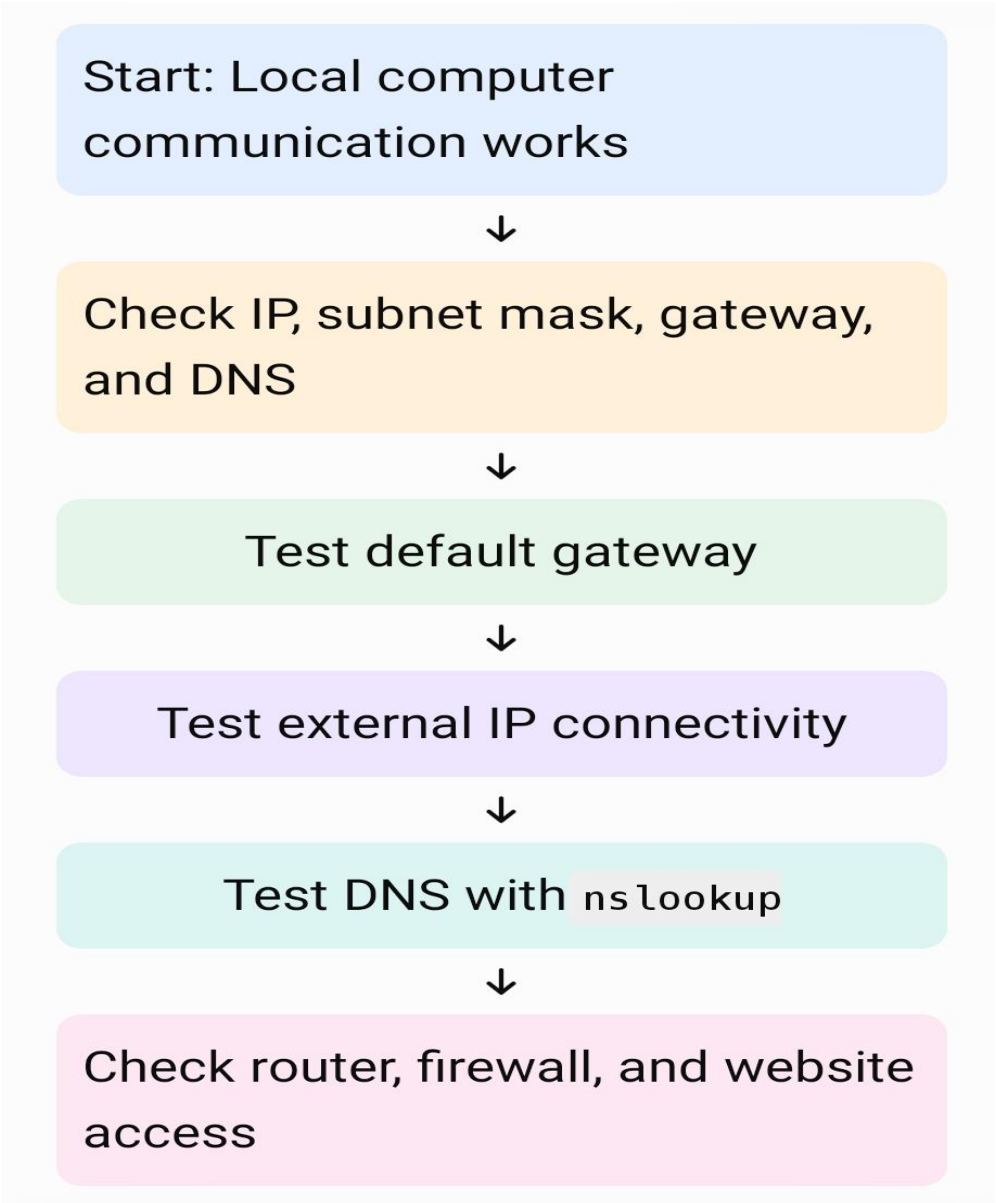
curl -I https://example.com

A response helps establish whether an HTTP connection can be made. If DNS and external connectivity work but the website does not, check proxy settings, firewall restrictions, TLS issues, and the website's availability.

Step 7: Verify the router's role

The router must have a working WAN connection and a valid route to the internet. Check its routing table, WAN status, NAT configuration where required, and firewall rules.

Troubleshooting flow diagram



Conclusion: Troubleshooting should proceed from local IP configuration to the gateway, external IP connectivity, DNS, and finally the router's internet connection and security settings. This systematic approach helps identify the cause of the problem.